

A PROJECT TITLE

ON

Credit Card Fraud Detection Using Logistic Regression, SVM and Isolation Forest

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

In

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by

Name SAI SURENDRANATH PENDYALA roll 2520030157

Name VANAPALLI ABHIDEV roll 2520030325

Under the Esteemed guidance of

Jyothsna Datti

Assistant Professor, CSE

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Credit Card Fraud Detection Using Logistic Regression, SVM and Isolation Forest

1. Abstract:

Credit card fraud is a major challenge in modern financial systems because fraudulent transactions can cause significant financial losses and reduce customer trust. This project, "Credit Card Fraud Detection Using Logistic Regression, SVM and Isolation Forest," aims to develop and compare multiple machine learning approaches for identifying fraudulent credit card transactions. The project will use the publicly available Credit Card Fraud Detection dataset, which contains European cardholder transactions, with a balanced distribution between legitimate and fraudulent transactions. The dataset will be preprocessed and divided into training and testing sets while applying appropriate techniques to handle class imbalance. Three approaches—Logistic Regression, Support Vector Machine (SVM), and Isolation Forest—will be implemented and compared. Logistic Regression provides an interpretable supervised classification baseline, SVM identifies complex decision boundaries between transaction classes, while Isolation Forest provides an unsupervised anomaly-detection approach suitable for identifying unusual transactions. The performance of the proposed models will be evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix, with particular emphasis on Precision, Recall, and F1-Score because fraud datasets are highly imbalanced. The final system will compare the strengths and limitations of each algorithm and determine which approach provides the most reliable fraud detection performance. The project aims to demonstrate how machine learning can support automated financial fraud detection while reducing false alarms and improving the identification of suspicious transactions.

2. INTRODUCTION:

The rapid growth of digital payments, online banking, and credit card transactions has made electronic financial services more convenient and accessible. However, this growth has also increased opportunities for fraudulent activities, including unauthorized transactions and stolen card usage. Detecting fraudulent transactions is challenging because fraudulent events usually represent only a very small proportion of all transactions. A model that simply predicts every transaction as legitimate may achieve high accuracy while failing to detect the transactions that matter most. Therefore, an effective fraud detection system must identify suspicious transactions while maintaining a reasonable balance between detecting fraud and minimizing false alarms.

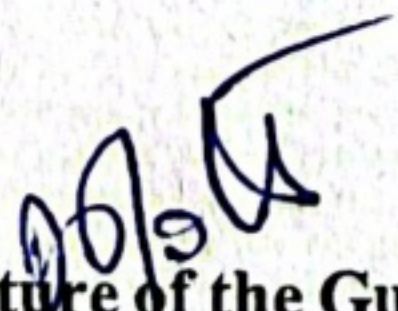
This project, "Credit Card Fraud Detection Using Logistic Regression, SVM and Isolation Forest," focuses on applying machine-learning techniques to automatically

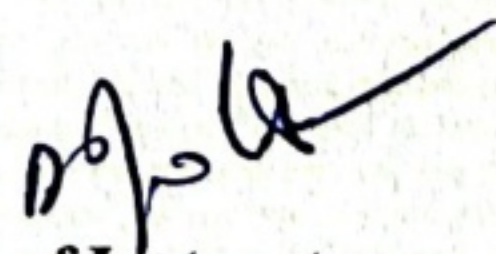
distinguish fraudulent transactions from legitimate ones. The project will use the Credit Card Fraud Detection dataset from the Kaggle. The dataset is balanced using the technique SMOTE(Synthetic Minority Over-sampling Technique). The dataset will be cleaned, normalized where necessary, and divided into suitable training and testing sets. Three different approaches will then be investigated: Logistic Regression as a supervised baseline classifier, Support Vector Machine (SVM) for learning effective classification boundaries, and Isolation Forest for detecting anomalous transactions without requiring fraud labels during anomaly detection. The models will be compared using Accuracy, Precision, Recall, F1-Score, ROC-AUC and Confusion Matrix.

3. SOFTWARE REQUIREMENTS:

The proposed fraud detection system requires a standard machine-learning development environment capable of performing data preprocessing, model training, evaluation, and visualization. Python will be used as the primary programming language, with Pandas and NumPy for data processing and Scikit-learn for implementing Logistic Regression, SVM, Isolation Forest, preprocessing, and evaluation metrics. Matplotlib and Seaborn can be used to visualize transaction distributions, confusion matrices, ROC curves, and model performance. The project can be developed using Visual Studio Code, PyCharm, Jupyter Notebook, or Google Colab.

Requirements	Specifications
Operating System	Windows 10/11 (64-bit) or Ubuntu Linux
Programming Language	Python 3.10 or above
IDE	Visual Studio Code / PyCharm / Jupyter Notebook / Google Colab
Machine Learning Library	Scikit-learn
Data Processing Library	Pandas
Numerical Computing Library	NumPy
Data Visualization	Matplotlib, Seaborn
Dataset	Credit Card Fraud Detection Dataset - Kaggle
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix
Algorithms	Logistic Regression, SVM, Isolation Forest
Data Storage	CSV
Version Control	Git & GitHub
Web Browser	Google Chrome / Microsoft Edge
Processor	Intel Core i5 and above
RAM	8 GB minimum, 16 GB recommended
Storage	256 GB SSD or above
GPU	Not required; optimal for experimentation


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